APMA 4302: Methods in Computational Science

Spring 2026

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Lecture 19

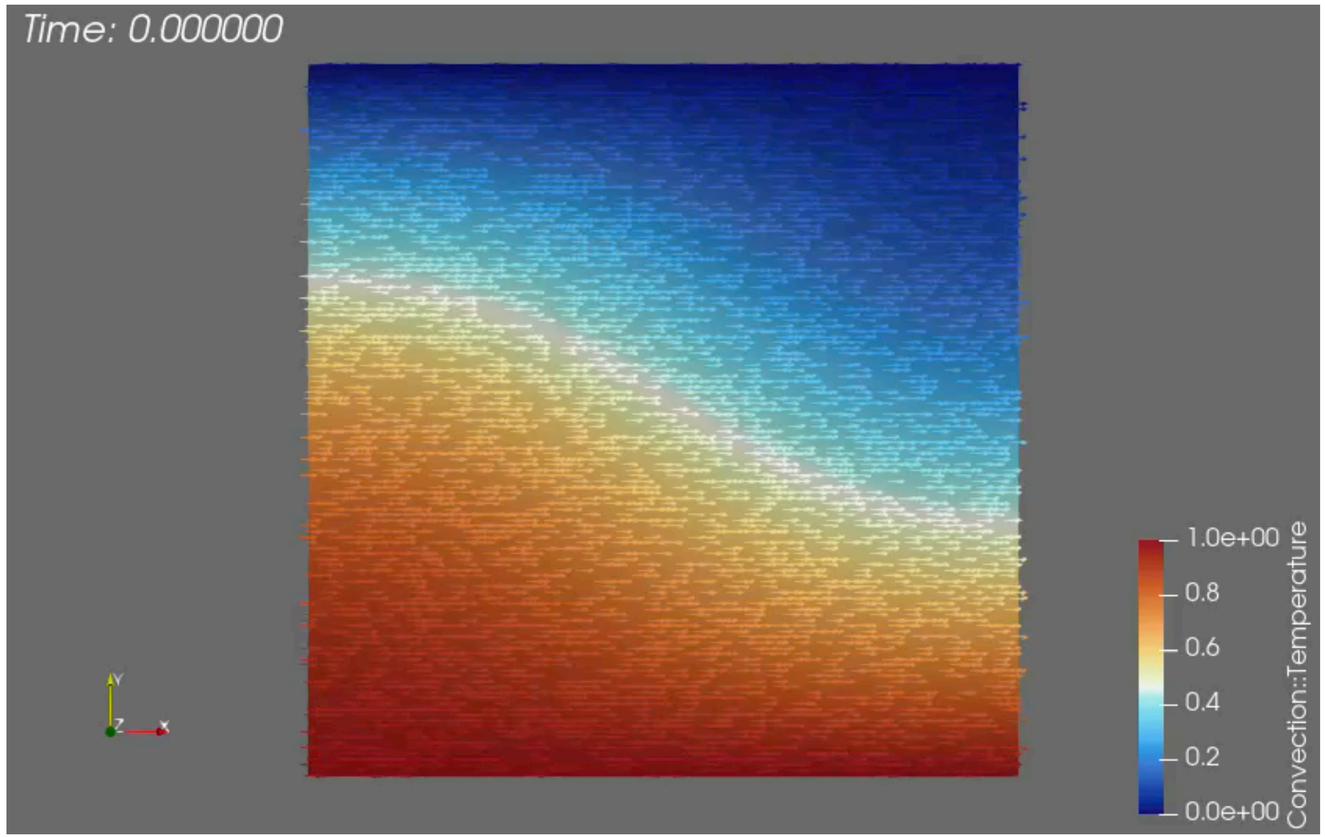
Advection-Diffusion Problems (Beuler Ch 11)

Outline

- Motivation and Challenges
- Basic physics: The Method of characteristics
- Finite Volume discretization
- Pure advection: 1-D and 2-D
- Advection + Diffusion: Arkimex schemes

Multi-physics problems

Towards effective multi-physics models



Advection-Diffusion Eq.

$$\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T = \frac{1}{Ra} \nabla^2 T$$

$$-\nabla \cdot \eta \left[(\nabla \mathbf{v}) + (\nabla \mathbf{v})^T \right] + \nabla P = T \hat{\mathbf{g}}$$

$$\nabla \cdot \mathbf{v} = 0$$

where $Ra = \dfrac{\rho_0 \alpha \Delta T h^3}{\eta \kappa}$ is the Raleigh #

and $\eta = \eta(T, \mathbf{v})$ is the viscosity

Incompressible Stokes flow

Problem #1: Infinite Prandtl number thermal convection

Other problems

Advection dominated multi-physics problems

- Navier Stokes/CFD
- Reactive Flow and Transport
- MHD/Plasma Physics
- Geophysical Fluid Dynamics
- Combustion
- Astrophysical fluid dynamics

Conservative and non-conservative forms

Advection-Diffusion Reaction Equations

Conservative form

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{F} = g(u, t, \mathbf{x})$$

with Flux

$$\mathbf{F} = \mathbf{v}u - \nu \nabla u + \ldots$$

Non-conservative form

$$\frac{\partial u}{\partial t} + \mathbf{v} \cdot \nabla u = \nabla \cdot \nu \nabla u + f(u, t, \mathbf{x})$$

pure advection

Solution by method of characteristics

$$\boxed{\frac{\partial u}{\partial t} + \mathbf{v} \cdot \nabla u = 0}$$

Is actually equivalent to a system of ODE's on particles

pure advection

Themes and variations: Source terms, non-linear problems

$$\boxed{\frac{\partial u}{\partial t} + \mathbf{v} \cdot \nabla u = g(u, t, \mathbf{x})}$$

can solve for $\mathbf{v}(\mathbf{x})$ or $\mathbf{v}(\mathbf{x}, u)$

pure advection

Finite Volume methods

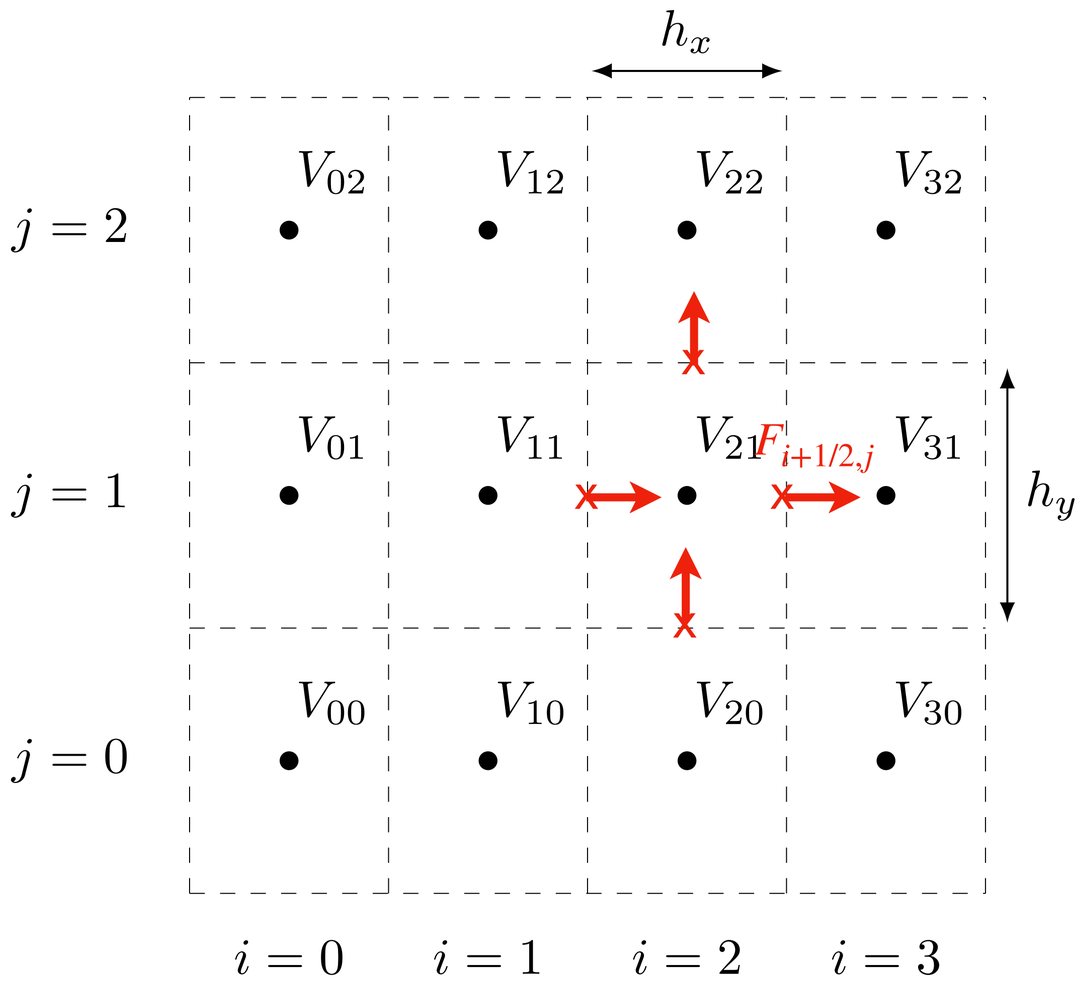
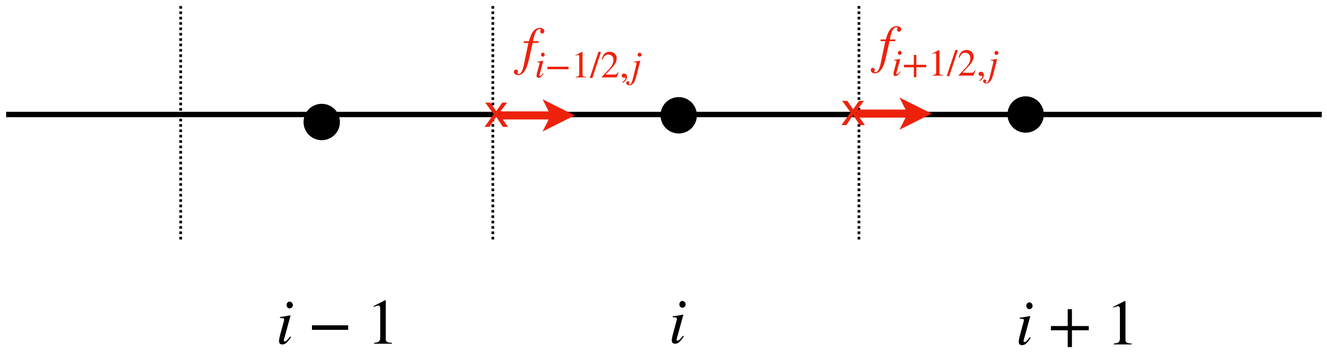


Figure 11.1. *Control volumes V_{ij} with cell centers at (x_i, y_j).*

pure advection

Finite Volume methods 1-D, advection

$$\frac{\partial u}{\partial t} + \frac{\partial f}{\partial x} = 0$$

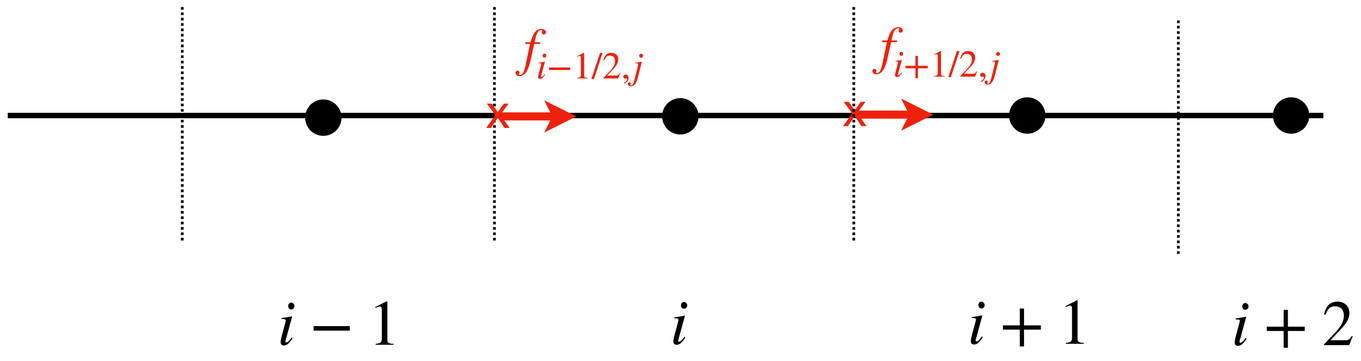


$$f = au$$

$$\frac{\partial u}{\partial t} + \frac{\partial au}{\partial x} = 0$$

pure advection

Finite Volume methods 1-D advection, choices for f



solve MOL system $$\dot{U}_i + \frac{f_{i+1/2} - f_{i-1/2}}{h} = 0$$

centered: $$f_{i+1/2} = a(x_{i+1/2})\left(U_{i+1} - U_i\right)$$

upwind: $$f_{i+1/2} = a^+(x_{i+1/2})U_i + a^-(x_{i+1/2})U_{i+1}$$

where: $$a^+ = \frac{a + |a|}{2} = \max(0,a) \qquad a^- = \frac{a - |a|}{2} = \min(0,a)$$

pure advection: issues

MOL methods become the linear ODE's $\dfrac{d\mathbf{U}}{dt} = A\mathbf{U}$ with Different A's

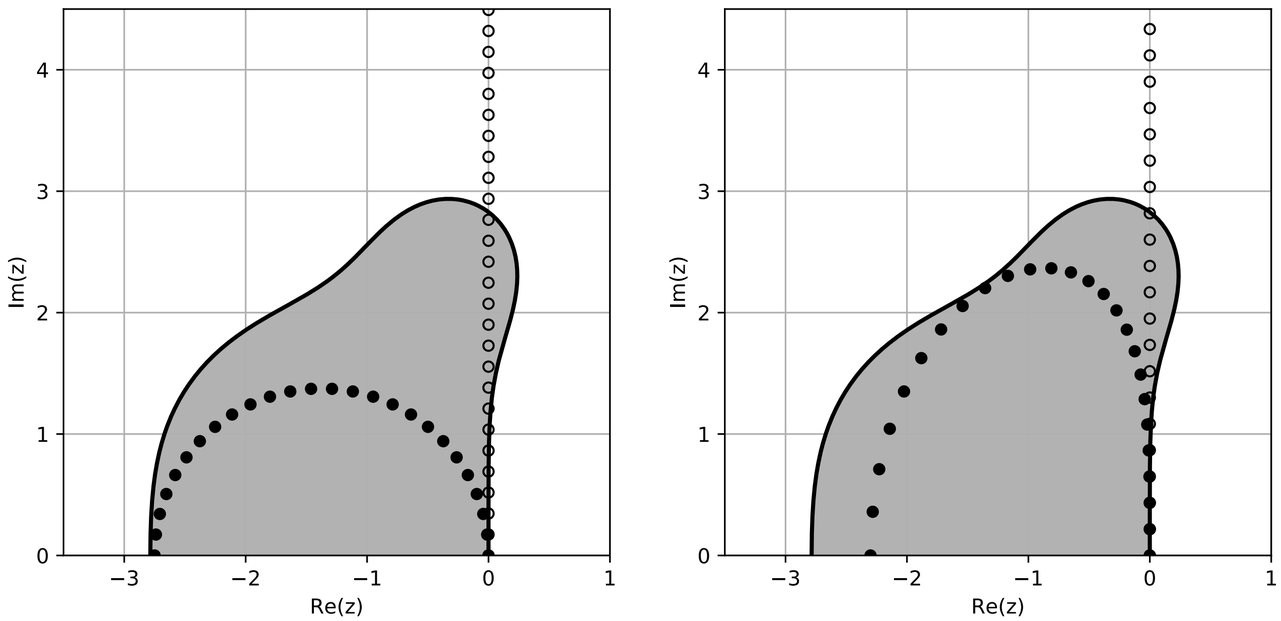


Figure 11.3. *The exact eigenvalues of PDE* (11.11) *(open circles) are imaginary. Those from first-order upwinding on the MOL system* (11.12) *(left; solid) or a third-order upwind-biased scheme* (11.22) *(right; solid) are damped by negative real parts. The absolute stability region of RK4 is shaded.*

Demonstration:

New PETSc code ts_advection_diffusion (in methods repo) solves

$$\frac{\partial u}{\partial t} + \frac{\partial au}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$$

using TS (with arkimex solvers), for constant a and analytic solution for a diffusing translating Gaussian

$$u(x,t) = \frac{\sigma}{\sqrt{\sigma^2 + 2\nu t}} \exp\left(-\frac{(x - x_0 - at)^2}{2(\sigma^2 + 2\nu t)} \right).$$

for different choices for f. for $\nu = 0$ is just pure advection

Fixes (?):

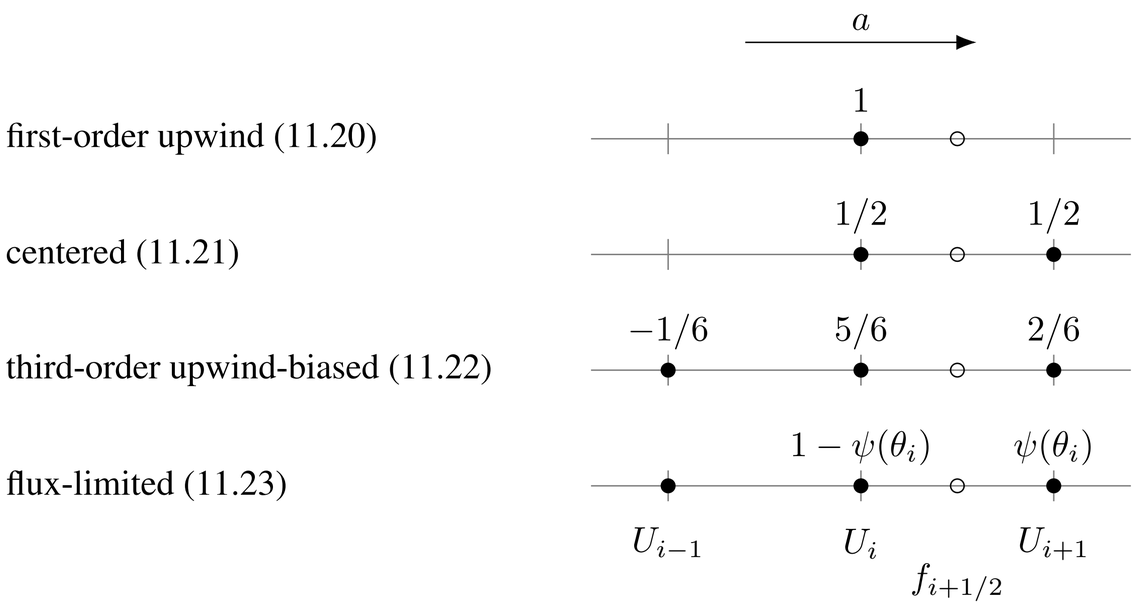
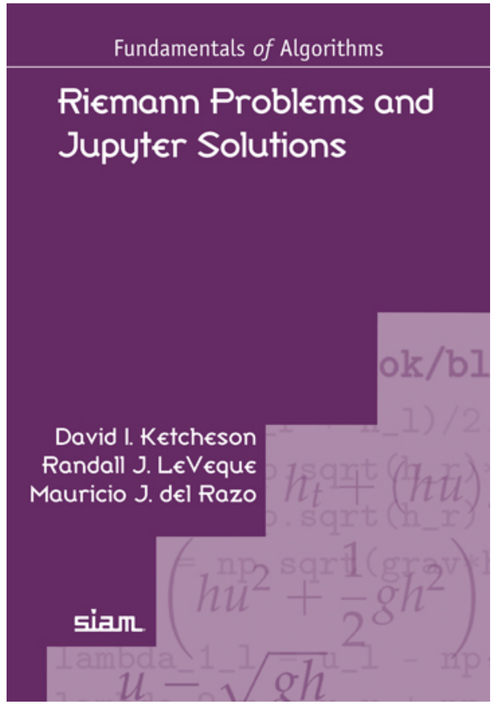


Figure 11.5. *When* $a = a(x_{i+1/2}) \geq 0$, *schemes* (11.23) *all use the same three regular grid points, with coefficients as shown.*

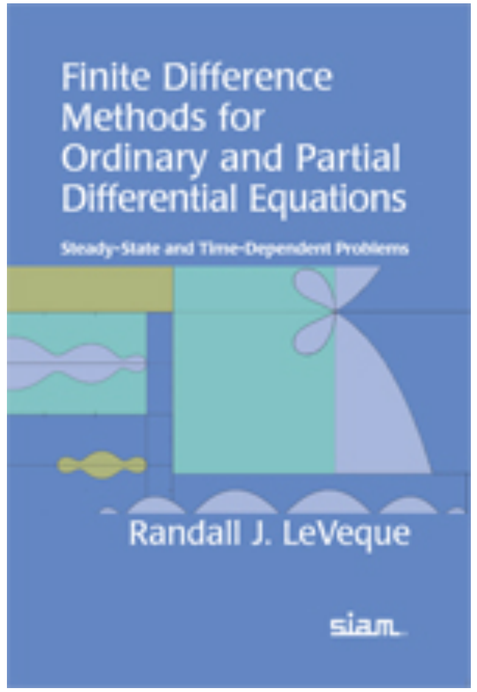
2-D demo: Beuler Ch11 p4pde's advect.c

References for Hyperbolic systems and PDEs

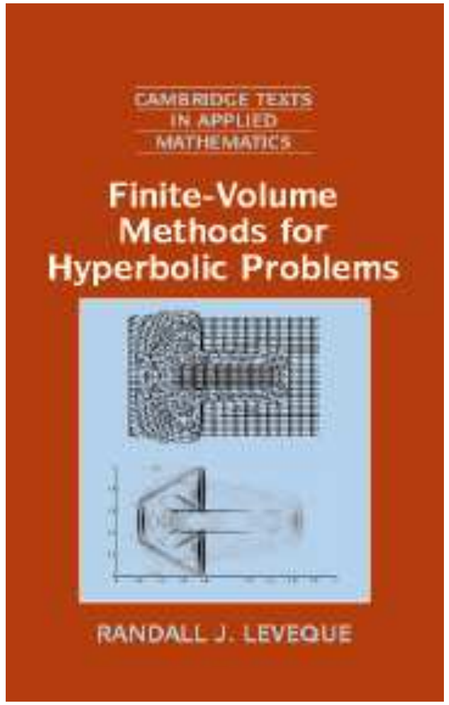
R. Leveque https://faculty.washington.edu/rjl/booksnotes.html



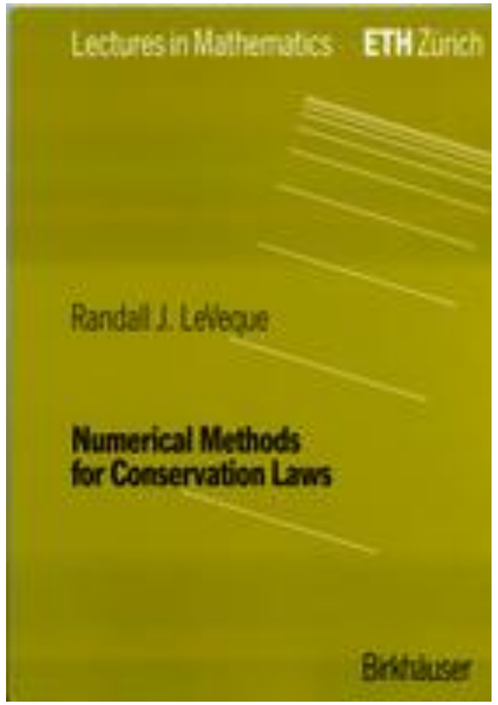
2020



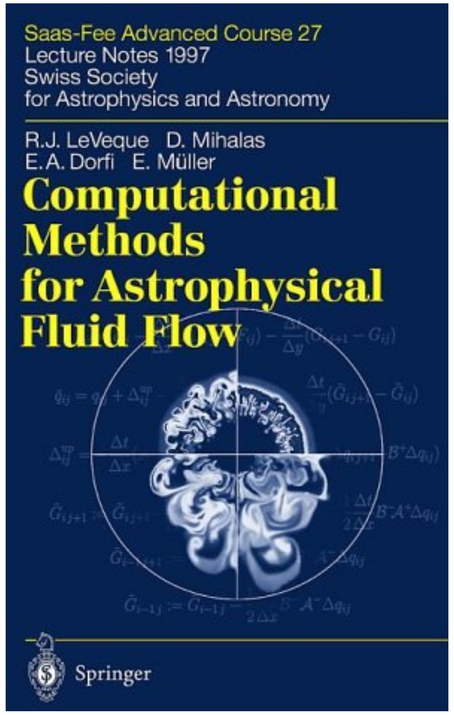
2007



2002



1990



1998

Combined Advection-Diffusion

TS Arkimex schemes